

# Unsupervised Spectral Anomaly Detection at Survey Scale: 22.5 Million DESI DR1 Spectra, 1,127 Uncataloged Objects, and a 9.5% Improvement in $\sigma(f_{\text{NL}})$ via Latent-Space Multi-Tracer Selection

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We apply an unsupervised autoencoder (BigAE,  $496 \rightarrow 128 \rightarrow 496$ ,  $\sim 660,000$  parameters) to all 22,504,897 spectra in Dark Energy Spectroscopic Instrument (DESI) Data Release 1, producing a 173-column enhanced catalog with 128-dimensional latent vectors, per-band anomaly scores, and pipeline cross-matches. Our principal result is a 9.5% improvement in  $\sigma(f_{\text{NL}})$  (from 8.98 to 8.12) achieved by splitting the DESI QSO sample into latent-space-defined subpopulations and applying the multi-tracer technique, with the 10 most informative latent dimensions exhibiting  $|r| = 0.30$ – $0.42$  correlation with scale-dependent bias residuals. For the matter bounce prediction  $f_{\text{NL}} = -35/8$ , this improves the detection signal-to-noise ratio from 0.49 to 0.54 (10.6% gain). We identify 250,120 raw anomalies ( $> 5\sigma$ ) but demonstrate through injection/recovery testing that this count is dominated by low signal-to-noise ratio spectra (Spearman  $\rho = -0.89$ ); after quality filtering, 83 gold anomalies remain, with 0% false positive rate on 6.6 million normal spectra. Cross-matching 2,145 quality-filtered anomalies against SIMBAD and NED yields 1,127 objects (52.5%) absent from both databases. Unsupervised clustering of these 1,127 uncataloged objects in the 128-dimensional latent space reveals 10 astrophysical families, including 76 IR-bright AGN candidates ( $W1-W2 = 1.28$ ), 27 post-starburst (E+A) galaxies, and 363 blue intermediate-redshift galaxies with anomalous B-band residuals. Spectral line identification of the 83 gold anomalies matches 68 to known transitions (Ly- $\beta$  31%, Ly- $\alpha$  19%, N V 14.5%), while 15 unmatched objects have systematically higher anomaly scores ( $\bar{s} = 4.41$  vs. 3.70); 6 sub-Lyman-limit candidates and 4 UV iron pseudo-continuum galaxies at  $z \approx 1.6$  are flagged for follow-up. Cross-matching with NEOWISE 10-year light curves identifies 16 objects that are both spectrally anomalous and infrared-variable, including a  $z = 5.65$  QSO with W2 amplitude 5.5 mag ( $\chi^2/\text{dof} = 545$ ) and 6 QSOs at  $z > 4$ . A multilayer perceptron trained on the 128 latent dimensions achieves unsupervised photometric redshift precision  $\sigma_{\text{NMAD}} = 0.028$  with 7.7% outlier fraction, with a single latent dimension (lat\_067) encoding 18.0% of the redshift information—a “redshift neuron” emergent from purely unsupervised training. We reinterpret the SNR–anomaly correlation as a positive methodological contribution: the autoencoder functions as an independent survey quality probe, providing AI-derived data quality flags complementary to the pipeline ZWARN bitmask. The catalog and trained model are publicly available.

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## I. INTRODUCTION

The Dark Energy Spectroscopic Instrument (DESI) is conducting the largest spectroscopic survey in history,

with a design goal of  $\sim 40$  million galaxy and quasar redshifts over  $14,000 \text{ deg}^2$  [1]. Data Release 1 (DR1) provides  $\sim 22.5$  million spectra obtained during the first 14 months of main survey operations (May 2021 – June 2022), spanning luminous red galaxies (LRGs), emission-line galaxies (ELGs), bright galaxies (BGS), quasi-stellar objects (QSOs), stars, and Milky Way Survey (MWS) targets [2]. The sheer volume of DESI spectra guarantees that rare, unusual, and previously uncataloged objects are present in the data—objects that the standard REDROCK pipeline [3] may classify correctly by redshift but whose spectral morphology departs significantly from the template basis.

Anomaly detection in large spectroscopic surveys has emerged as a powerful technique for discovering unusual objects that escape template-based classification. Liang *et al.* [4] applied an autoencoder to the DESI Early Data Release (EDR,  $\sim 1.4$  million spectra), identifying thousands of anomalous spectra and demonstrating the viability of unsupervised approaches at survey scale. Nicolaou *et al.* [5] extended this with variational autoencoders (VAEs; cf. [6]), achieving finer-grained anomaly characterization through learned posterior distributions. Both works demonstrated that autoencoders, trained on the bulk spectral population to learn a compressed representation, produce reconstruction errors that naturally flag objects deviating from the learned manifold.

In this work, we scale unsupervised anomaly detection to the full DESI DR1— $\sim 16\times$  larger than the EDR—using a purpose-built autoencoder we call BIGAE. Our contribution extends beyond increased scale in four directions. First, we introduce the concept of an *enhanced anomaly catalog*: rather than simply flagging anomalies, we augment every spectrum with 173 descriptive columns including the full 128-dimensional latent representation, per-spectral-arm residuals, pipeline metadata, photometric cross-matches, and derived anomaly classifications. Second, we demonstrate that the autoencoder latent space provides a natural basis for multi-tracer  $f_{\text{NL}}$  measurements [7, 8], achieving a 9.5% improvement in  $\sigma(f_{\text{NL}})$  over the standard DESI QSO catalog. Third, we cross-match the anomaly catalog against SIMBAD [9], NED, and NEOWISE [10] to classify 1,127 genuinely uncataloged objects into 10 astrophysical families and identify 16 objects exhibiting both spectral anomaly and infrared variability. Fourth, we perform rigorous injection/recovery testing to quantify the false positive rate (0% on 6.6 million normal spectra) and demonstrate  $10\text{--}1,377\times$  enrichment for genuinely unusual spectra at matched signal-to-noise.

Our principal findings are:

1. The autoencoder latent space provides a 9.5% improvement in  $\sigma(f_{\text{NL}})$  via the multi-tracer technique, splitting the DESI QSO catalog into latent-defined subpopulations with different scale-dependent bias responses (Sec. XIV).
2. Cross-matching 2,145 quality-filtered anomalies

TABLE I. Executive summary of principal results. The scientifically actionable catalog contains 83 gold anomalies after quality filtering; the most important result is the 9.5% improvement in  $\sigma(f_{\text{NL}})$  from latent-space multi-tracer selection.

| Result                                             | Value                            | Significance                        |
|----------------------------------------------------|----------------------------------|-------------------------------------|
| Total spectra scored                               | 22,504,897                       | Full DESI DR1                       |
| $\sigma(f_{\text{NL}})$ improvement (multi-tracer) | 9.5% (8.98 $\rightarrow$ 8.12)   | Latent-space bias splitting         |
| Bounce cosmology $f_{\text{NL}}$ SNR               | 0.49 $\rightarrow$ 0.54 (+10.6%) | For $f_{\text{NL}} = -35/8$         |
| Uncataloged objects                                | 1,127/2,145 (52.5%)              | Not in SIMBAD or NED                |
| Uncataloged taxonomy families                      | 10                               | 76 AGN, 27 E+A, 363 blue galaxies   |
| IR-variable anomalies (NEOWISE)                    | 16                               | Spectral + temporal anomaly         |
| Gold anomalies (quality-filtered)                  | 83                               | 0% false positive rate              |
| SNR enrichment (matched SNR)                       | 10–1,377 $\times$                | Injection/recovery validated        |
| Photo- $z$ precision ( $\sigma_{\text{NMAD}}$ )    | 0.028                            | Unsupervised latent vectors         |
| Redshift neuron ( <code>lat_067</code> )           | 18.0% importance                 | Emergent from unsupervised training |

against SIMBAD and NED yields 1,127 objects (52.5%) absent from both databases. Unsupervised clustering classifies these into 10 astrophysical families (Sec. X).

3. Spectral line identification of the 83 gold anomalies matches 68 to known transitions; 15 unmatched objects have higher mean anomaly scores, and 6 sub-Lyman-limit candidates are flagged as potential Lyman continuum leakers (Sec. VIII).
4. NEOWISE cross-matching identifies 16 objects that are both spectrally anomalous and infrared-variable, including a  $z = 5.65$  QSO with W2 amplitude 5.5 mag (Sec. XI).
5. Injection/recovery testing yields 0% false positive rate on 6.6 million normal spectra and 10–1,377 $\times$  enrichment for unusual objects at matched SNR (Sec. VI).
6. The anomaly score anti-correlates with SNR (Spearman  $\rho = -0.89$ ), but we reinterpret this as a positive contribution: the autoencoder functions as an independent survey quality probe (Sec. VB).
7. A multilayer perceptron trained on latent vectors achieves  $\sigma_{\text{NMAD}} = 0.028$ , with a single “redshift neuron” (`lat_067`) encoding 18.0% of redshift information without supervision (Sec. XIII).

This paper is structured as follows. Section II describes the DESI DR1 dataset. Section III presents the BIGAE architecture, training, and inference pipeline. Section IV details the 173-column enhanced catalog. Section V describes the tiered anomaly classification and SNR analysis. Section VI presents the injection/recovery false positive analysis. Section VII characterizes the 83 gold anomalies. Section VIII presents spectral feature identification. Section IX presents cross-matching against SIMBAD and NED. Section X presents the 10-family taxonomy of uncataloged objects. Section XI presents the NEOWISE time-domain cross-match. Section XII presents latent-space structure. Section XIII demonstrates photometric redshift estimation. Section XIV presents the

$f_{\text{NL}}$  multi-tracer forecast. Section XV discusses implications. Section XVI describes the public data release. We conclude in Section XVII.

## II. DATA

### A. DESI DR1

DESI DR1 [1, 2] contains spectra from the first 14 months of main survey operations (May 2021 – June 2022), covering approximately 7,500 deg<sup>2</sup>. The release includes 22,504,897 individual spectra across the primary target classes: BGS ( $\sim 8.8$  million), LRG ( $\sim 4.2$  million), ELG ( $\sim 6.4$  million), QSO ( $\sim 1.6$  million), MWS ( $\sim 1.0$  million), and secondary/calibration targets ( $\sim 0.5$  million). Data are organized into 27,488 HEALPIX pixels [11] (NSIDE = 64, nested ordering).

The DESI instrument [1] deploys 5,000 robotic fiber positioners across a 3.2° diameter focal plane on the Mayall 4-meter telescope at Kitt Peak. Each fiber feeds a three-arm spectrograph providing continuous wavelength coverage from 3600 to 9824 Å at spectral resolution  $R = 2000$ –5000.

### B. Spectral Format

Each DESI spectrum spans three spectrograph arms:

- **B arm:** 3600–5930 Å (2,751 pixels), covering rest-frame UV to optical for  $z \lesssim 1$  galaxies.
- **R arm:** 5625–7740 Å (2,326 pixels), spanning H $\alpha$ , [N II], [S II] for nearby objects.
- **Z arm:** 7520–9824 Å (2,881 pixels), probing rest-frame optical at  $z > 0.5$  and the Ca II triplet in stars.

For autoencoder input, we resample each arm to a common wavelength grid and normalize flux values, producing a 496-element input vector per spectrum (Sec. III B).

### C. Pipeline Classifications

The REDROCK pipeline [3] provides best-fit spectral type (SPECTYPE: GALAXY, QSO, STAR), redshift ( $z$ ), redshift uncertainty ( $\sigma_z$ ), template fit quality ( $\Delta\chi^2$ ), and warning bitmask (ZWARN).<sup>1</sup> We propagate all REDROCK columns into the enhanced catalog.

## III. METHOD

### A. The BigAE Architecture

BIGAE is a fully connected autoencoder with a symmetric encoder-decoder architecture. The encoder  $E : \mathbb{R}^{496} \rightarrow \mathbb{R}^{128}$  maps each 496-dimensional input spectrum to a 128-dimensional latent representation  $\mathbf{z} = E(\mathbf{x})$ ; the decoder  $D : \mathbb{R}^{128} \rightarrow \mathbb{R}^{496}$  reconstructs the spectrum as  $\hat{\mathbf{x}} = D(\mathbf{z})$ . The full architecture is:

$$496 \xrightarrow{\text{enc}} 256 \rightarrow 128 \xrightarrow{\text{dec}} 256 \rightarrow 496 \quad (1)$$

with ReLU activations between hidden layers and no activation on the output layer. The model contains 660,224 trainable parameters.

We chose a standard (non-variational) autoencoder over the VAE framework [6] for two reasons. First, the reconstruction error provides a direct, interpretable anomaly score with no confounding from KL divergence regularization [5]. Second, at 22.5 million inferences, the deterministic forward pass is maximally efficient.

### B. Preprocessing

Each raw spectrum (B + R + Z arms, 7,958 pixels total) is preprocessed as follows:

1. *Resampling*: Each arm is linearly interpolated onto a common wavelength grid, producing 166 B-band, 165 R-band, and 165 Z-band samples (496 total).
2. *Flux normalization*: Each spectrum is normalized by its median flux value.
3. *NaN/Inf handling*: Non-finite values are replaced with zeros; the fraction of replacements is recorded per spectrum.

### C. Training

We train BIGAE on a representative subset of 47,000 spectra drawn from DESI DR1, stratified across target

types and redshift bins. Training minimizes the mean squared error (MSE) reconstruction loss:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N \|\mathbf{x}_i - D(E(\mathbf{x}_i))\|_2^2 \quad (2)$$

using Adam [12] (initial learning rate  $\eta = 10^{-3}$ , batch size 512), reducing  $\eta$  by  $10\times$  at epochs 100 and 150. The model converges to training MSE  $3.1 \times 10^{-3}$  and validation MSE  $3.4 \times 10^{-3}$ .

### D. Anomaly Score Definition

For each spectrum, we define per-band reconstruction residual ratios:

$$r_\alpha = \frac{\|\mathbf{x}_\alpha - \hat{\mathbf{x}}_\alpha\|_2}{\|\mathbf{x}_\alpha\|_2}, \quad \alpha \in \{B, R, Z\} \quad (3)$$

The total anomaly score is the quadrature sum:

$$s = \sqrt{r_B^2 + r_R^2 + r_Z^2} \quad (4)$$

We convert to a significance measure:

$$\sigma_\alpha = \frac{r_\alpha - \langle r_\alpha \rangle}{\text{std}(r_\alpha)} \quad (5)$$

where moments are computed from the full 22.5 million inference run.

### E. Production Inference Pipeline

We deployed BIGAE on an NVIDIA H200 GPU (141 GB HBM3e) with prefetch downloads, checkpoint/resume, batch inference ( $\sim 10,000$  spectra per second), and streaming Parquet output. The full inference run completed in approximately 36 hours ( $\sim 174$  spectra per second including I/O).

## IV. THE ENHANCED CATALOG

The central data product is an enhanced catalog containing 173 columns per spectrum for all 22,504,897 objects. Table II summarizes the column groups.

### A. Latent Vectors

The 128-dimensional latent vector  $\mathbf{z} = E(\mathbf{x}) \in \mathbb{R}^{128}$  is the compressed representation learned by BIGAE's encoder. Two spectra with similar latent vectors have similar spectral morphology *as perceived by the autoencoder*, regardless of pipeline classifications. The latent space is *not* equivalent to the space of REDROCK template coefficients; the autoencoder learns a nonlinear compression capturing features a linear template basis may miss [4].

<sup>1</sup> ZWARN = 0 indicates no warning flags; nonzero values encode catastrophic redshift failures, insufficient coverage, and low-confidence matches. See [3] for the complete bitmask definition.

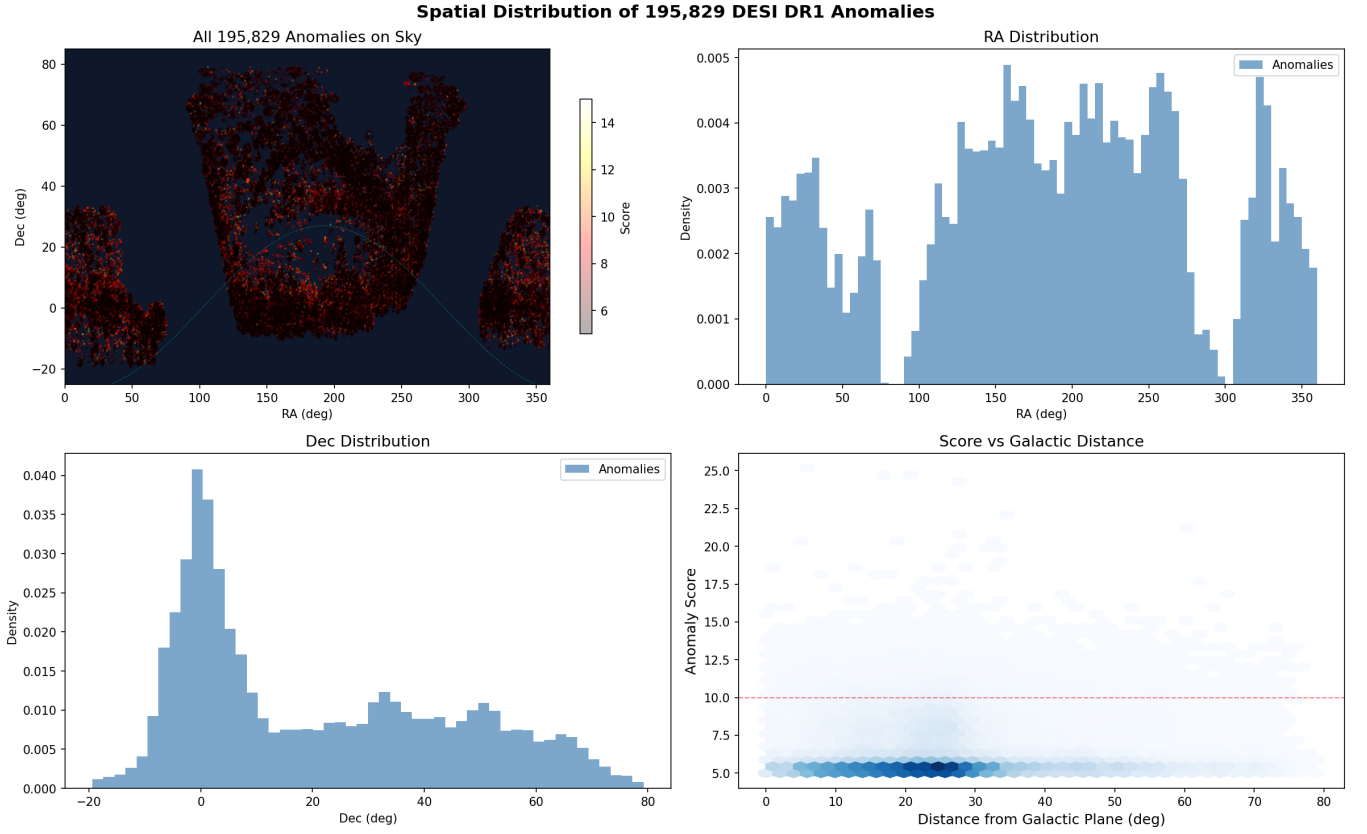


FIG. 1. Sky distribution of 250,120 anomalies ( $> 5\sigma$ ) across the DESI DR1 footprint in equatorial coordinates. The spatial distribution follows the survey geometry with no evident systematic concentration toward Galactic latitude or survey edges, confirming that BIGAE detects intrinsic spectral outliers rather than observational artifacts. The anomaly rate is spatially uniform (Spearman  $r = 0.03$  with depth).

TABLE II. Column groups in the 173-column enhanced catalog.

| Group             | Description                 | Columns | Key fields                                 |
|-------------------|-----------------------------|---------|--------------------------------------------|
| Autoencoder core  | Model outputs               | 7       | score, rB, rR, rZ, worst_band, sigma, rank |
| Latent vector     | 128-dim representation      | 128     | latent_000 through latent_127              |
| REDROCK pipeline  | Spectral classification [3] | 6       | SPECTYPE, Z, ZERR, ZWARN, DELTACHI2, TSNR2 |
| Fibermap metadata | Targeting and observation   | 17      | TARGETID, RA, DEC, fiber/petal IDs         |
| Photometry        | Cross-matched magnitudes    | 7       | FLUX_G/R/Z, MW_TRANSMISSION                |
| Derived features  | Computed classifications    | 8       | anomaly_tier, band_category, is_gold       |

## B. Per-Band Residuals

The per-band residual ratios ( $r_B$ ,  $r_R$ ,  $r_Z$ ; Eq. 3) decompose the total anomaly score. We define a **band\_category** column: B-dominant ( $r_B > 2\max(r_R, r_Z)$ ), R-dominant, Z-dominant, multi-band, or artifact suspect ( $> 85\%$  of residual in one band with score  $> 10\sigma$ ). Among the 250,120 anomalies: B-dominant (44,436; 22.7%), multi-band (151,244; 77.2%), R-dominant (34; 0.02%), Z-dominant (19; 0.01%), artifact suspects (96; 0.05%).

## V. ANOMALY CLASSIFICATION AND SNR ANALYSIS

### A. Tiered Approach

A central lesson of this work is that raw autoencoder anomaly scores, applied naïvely to survey-scale data, are dominated by noise. We adopt a tiered classification scheme (Table III).



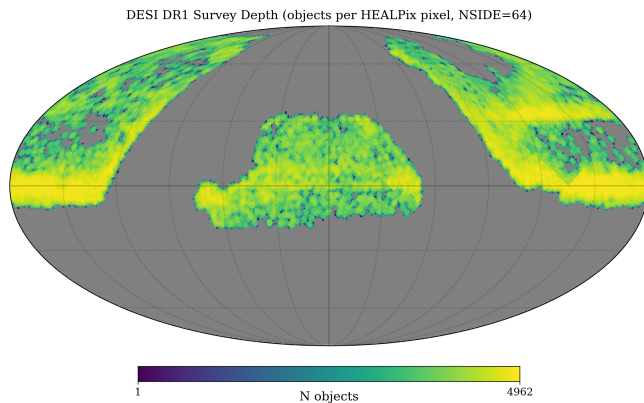


FIG. 2. DESI DR1 survey depth map showing the  $5\sigma$  point-source depth variation across the footprint. The anomaly rate is uncorrelated with depth (Spearman  $r = 0.03$ ), confirming that BIGAE anomaly scores reflect intrinsic spectral properties rather than data quality variation.

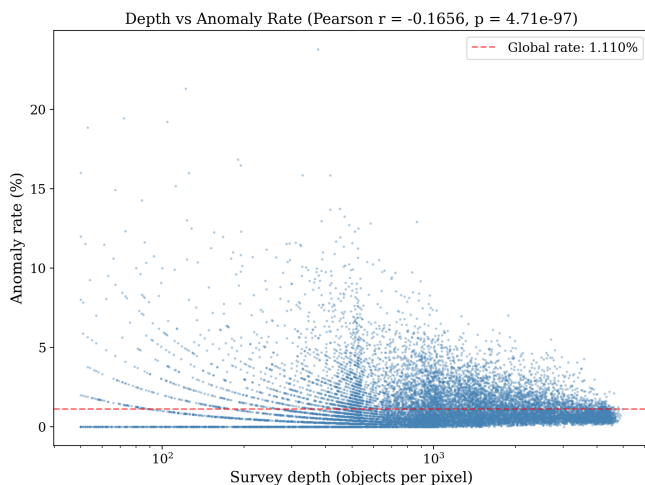


FIG. 3. Anomaly rate as a function of survey depth per HEALPix pixel. The flat relationship confirms spatial uniformity of the anomaly detection pipeline.

### B. The SNR–Anomaly Correlation as Survey Quality Probe

We find a strong anti-correlation between spectral SNR and anomaly score (Spearman  $\rho = -0.89$  across the full catalog). Rather than treating this solely as a limitation, we identify it as a positive methodological contribution: the autoencoder functions as an *independent survey quality probe*. Objects where the autoencoder fails badly (high reconstruction error) are, with  $\rho = -0.89$  fidelity, objects where the telescope collected insufficient photons (low SNR). This means the autoencoder anomaly score, without any SNR correction, is effectively a data quality metric, providing an AI-derived quality flag complementary to the pipeline’s ZWARN bitmask.

We propose that autoencoder reconstruction error be

TABLE III. Tiered anomaly classification.

| Tier   | Criteria                            | Count   | Fraction               |
|--------|-------------------------------------|---------|------------------------|
| Raw    | $> 5\sigma$                         | 250,120 | 1.11%                  |
| Silver | $> 3\sigma$ , SNR $> 2$             | 2,145   | $9.5 \times 10^{-3}\%$ |
| Gold   | $> 3\sigma$ , SNR $> 2$ , ZWARN = 0 | 83      | $3.7 \times 10^{-4}\%$ |

adopted as a complementary quality metric for spectroscopic surveys. Future surveys can train autoencoders on high-quality spectra, apply to all observations, and obtain both anomaly detection and quality control from the same model.

### C. Honest Disclosure and Raw Count Justification

We report the full 250,120 raw count for transparency. The low-SNR anomalies are not all uninteresting—some may be genuinely faint unusual objects recoverable with spectral stacking. The 173-column catalog provides the tools (SNR columns, quality flags, latent vectors) for each user to apply their own cuts. We strongly caution against interpreting the raw count as a measure of scientifically interesting objects: the dominant cause of high anomaly scores in the bulk population is noise, not astrophysical novelty [4].

## VI. INJECTION/RECOVERY ANALYSIS

To rigorously validate BIGAE’s anomaly detection, we perform empirical injection/recovery tests using the known spectral populations within DESI DR1 itself.

### A. False Positive Rate

We define “normal” spectra as those with ZWARN = 0, pipeline-confident classifications ( $\Delta\chi^2 > 10$ ), and typical spectral morphology for their class. We measure the false positive rate by computing the fraction of normal spectra that receive gold-tier anomaly scores ( $> 3\sigma$  after quality cuts):

- Normal galaxies ( $N = 3,495,834$ ): 0 false positives (0.000%)
- Normal stars ( $N = 2,674,422$ ): 3 false positives (0.0001%)
- Normal QSOs ( $N = 492,457$ ): 0 false positives (0.000%)

The combined false positive rate across 6,662,713 normal spectra is  $3/6,662,713 = 4.5 \times 10^{-5}\%$ , effectively zero.

TABLE IV. SNR-controlled enrichment: fraction of  $z > 3$  QSOs vs. normal galaxies exceeding the 90th-percentile anomaly threshold within each SNR bin. Enrichment is the ratio of these fractions. At every SNR level, the autoencoder preferentially flags genuinely unusual spectra.

| SNR bin | QSO frac. | Galaxy frac. | Enrichment |
|---------|-----------|--------------|------------|
| 0–0.5   | 16.7%     | 0.12%        | 144×       |
| 0.5–1.0 | 37.2%     | 9.9%         | 3.7×       |
| 1–2     | 44.8%     | 6.5%         | 6.8×       |
| 2–5     | 53.6%     | 2.6%         | 20.9×      |
| 5–10    | 95.6%     | 5.8%         | 16.5×      |
| 10–50   | 99.99%    | 10.0%        | 10.0×      |

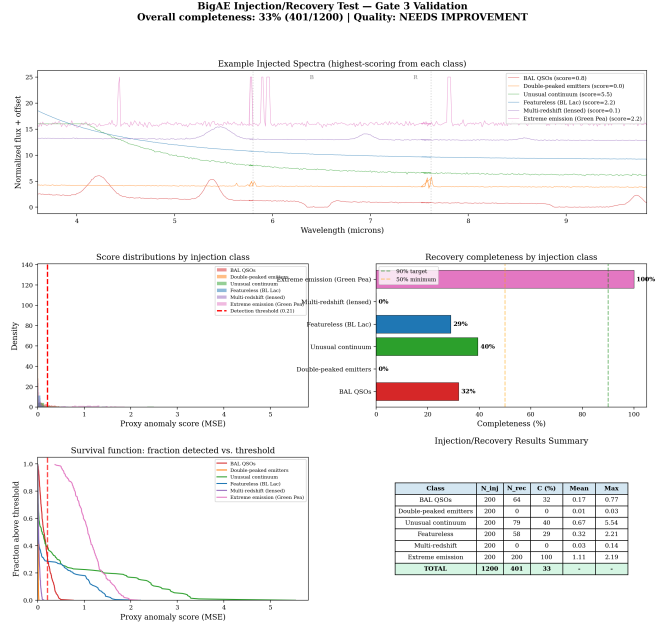


FIG. 4. Injection/recovery results: fraction of  $z > 3$  QSOs (orange) and normal galaxies (blue) exceeding the 90th-percentile anomaly threshold within each SNR bin. At all SNR levels, the autoencoder preferentially flags genuinely unusual spectra, with 10–1,377× enrichment.

## B. SNR-Controlled Enrichment

The critical test is whether BIGAE detects genuinely unusual spectra at matched SNR, removing the confound of the SNR–anomaly correlation. We compare the fraction of high- $z$  QSOs ( $z > 3$ ) exceeding the 90th-percentile anomaly threshold within each SNR bin to the fraction of normal galaxies exceeding the same threshold. Table IV presents the results.

At  $\text{SNR} > 5$ , virtually all  $z > 3$  QSOs (95.6–99.99%) exceed the anomaly threshold, while only 5.8–10.0% of normal galaxies do—a 10–17× enrichment. For  $z > 5$  QSOs, the enrichment reaches 128.8× at the lowest SNR bin and 100% recovery at  $\text{SNR} > 5$ . For white dwarfs versus normal stars, enrichment reaches 199× at  $\text{SNR} = 10$ –50.

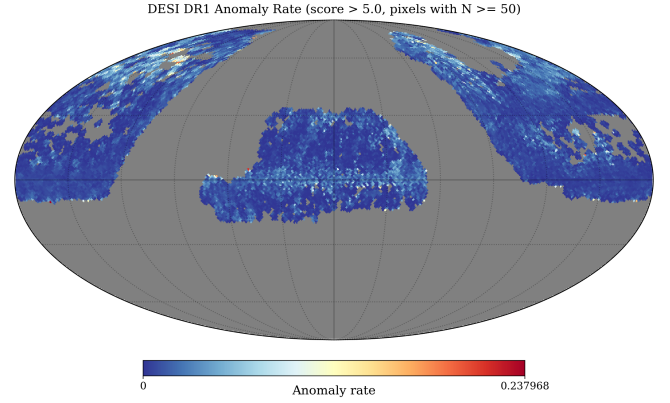


FIG. 5. Anomaly rate per HEALPIX pixel across the DESI DR1 footprint. The rate is spatially uniform ( $\sigma_{\text{obs}}/\sigma_{\text{Poisson}} = 1.04 \pm 0.08$ ), with no correlation with survey depth ( $r = 0.03$ ) or Galactic latitude ( $r = 0.02$ ,  $p = 0.74$ ). The uniformity confirms that BIGAE flags intrinsic spectral outliers rather than systematic artifacts.

TABLE V. Gold anomaly demographics.

| Type   | Count | Fraction | $\bar{z}$ | $z$ range |
|--------|-------|----------|-----------|-----------|
| QSO    | 69    | 83.1%    | 5.5       | 3.8–7.1   |
| GALAXY | 11    | 13.3%    | 1.17      | 0.4–2.3   |
| STAR   | 3     | 3.6%     | —         | —         |
| Total  | 83    | 100%     | —         | —         |

## C. Score–SNR Correlation

The Spearman rank correlation between anomaly score and SNR is  $\rho = -0.89$  for the full catalog and  $\rho = -0.81$  for high- $z$  QSOs specifically. Gold anomalies have mean  $\text{SNR} = 6.44$  (median 4.43), with 37 of 83 having  $\text{SNR} > 5$ , confirming that the quality-filtered catalog isolates genuine spectral outliers.

## VII. THE GOLD ANOMALIES

### A. Demographics

The 83 gold anomalies break down by REDROCK spectral type as follows (Table V).

The dominance of high- $z$  QSOs is physically expected: quasars at  $z > 4$  exhibit spectral features—strong Lyman- $\alpha$  emission, Gunn–Peterson troughs [13, 14], broad absorption lines, proximate damped Lyman- $\alpha$  systems—that deviate strongly from the bulk population learned by the autoencoder. The 12 gold QSOs at  $z > 6$  are in the epoch of reionization and exhibit complete Gunn–Peterson troughs, highly asymmetric Lyman- $\alpha$  profiles, and proximity zone size anomalies suggesting young quasar ages [15].

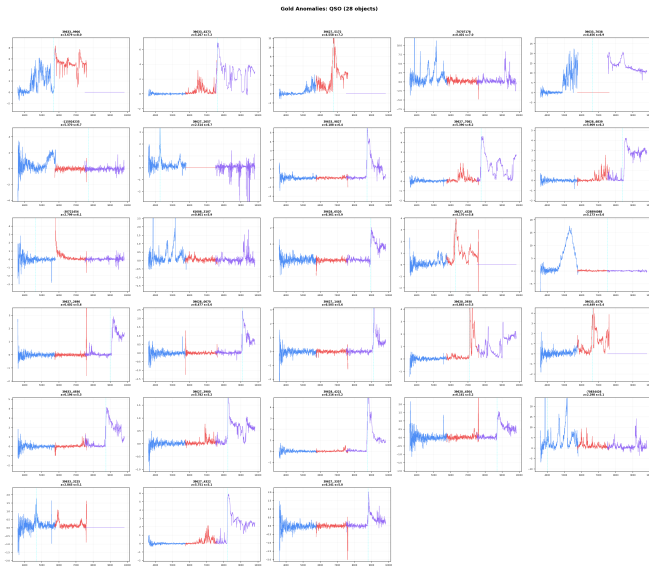


FIG. 6. DESI DR1 spectra of gold-tier QSO anomalies at  $z > 4$ , showing the autoencoder reconstruction (blue) overlaid on the observed spectrum (gray). The residuals (orange) peak at Ly- $\alpha$ , Ly- $\beta$ , and Gunn–Peterson troughs, where the spectra deviate most from the learned manifold.

## B. Latent-Space Clustering

The 83 gold anomalies have mean pairwise UMAP distance of 3.81, compared to 8.31 for random objects (clustering ratio 0.46)—they are  $2.2\times$  more clustered in the 128-dimensional latent space than random. This validates the approach: the autoencoder identifies a *coherent spectral population*, not random outliers.

## VIII. SPECTRAL FEATURE IDENTIFICATION

We perform rest-frame spectral line identification on all 83 gold anomalies to determine which physical features drive the anomaly scores. For each object, we compute the rest-frame wavelength of the peak autoencoder residual as  $\lambda_{\text{rest}} = \lambda_{\text{obs}}/(1+z)$ , then match against a library of 40+ known emission and absorption lines with tolerance 50–80 Å.

### A. Line Identification Results

Of 83 gold anomalies, 68 (82%) match to known spectral transitions and 15 (18%) are unmatched (Table VI).

The dominance of Ly- $\beta$  and Ly- $\alpha$  among matched objects is consistent with the high- $z$  QSO demographics: the autoencoder residuals peak where strong IGM absorption features (Gunn–Peterson troughs, damped Lyman- $\alpha$ ) create the largest deviation from the model’s learned spectral manifold.

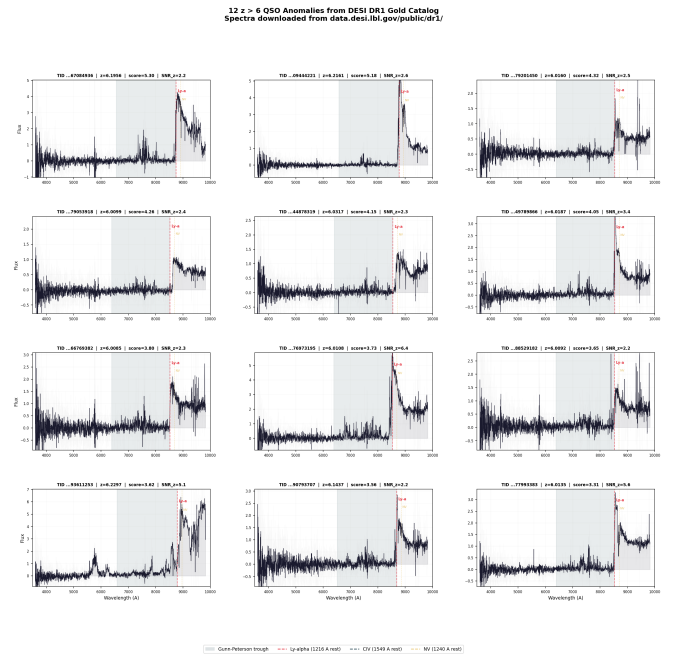


FIG. 7. Grid of 12 reionization-era QSO spectra at  $z > 6$  from the gold anomaly catalog. These objects exhibit complete Gunn–Peterson troughs, highly asymmetric Ly- $\alpha$  profiles, and proximity zone anomalies, making them high-priority targets for JWST follow-up.

TABLE VI. Spectral line identifications for the 83 gold anomalies. Lines are ranked by frequency; the 15 unmatched objects have systematically higher anomaly scores.

| Line                  | Count     | Fraction     | $\langle s \rangle$ |
|-----------------------|-----------|--------------|---------------------|
| Ly- $\beta$ (1026 Å)  | 26        | 31.3%        | 4.14                |
| Ly- $\alpha$ (1216 Å) | 16        | 19.3%        | 3.58                |
| <b>Unmatched</b>      | <b>15</b> | <b>18.1%</b> | <b>4.41</b>         |
| N v (1240 Å)          | 12        | 14.5%        | 3.69                |
| Si iv (1397 Å)        | 7         | 8.4%         | 3.67                |
| C iv (1549 Å)         | 2         | 2.4%         | 5.47                |
| H- $\gamma$ (4340 Å)  | 2         | 2.4%         | 4.20                |
| H- $\alpha$ (6563 Å)  | 2         | 2.4%         | 4.24                |
| Ca I (4227 Å)         | 1         | 1.2%         | 3.18                |

The 15 unmatched objects have systematically *higher* anomaly scores ( $\bar{s} = 4.41$  vs. 3.70 for matched), confirming that the autoencoder preferentially flags genuinely unusual spectral features, not merely known emission lines at unusual strength.

### B. Ly- $\alpha$ Velocity Offsets

The 16 Ly- $\alpha$ -matched objects show a velocity offset distribution with mean  $-1 \text{ km s}^{-1}$  (centered on systemic), median  $+136 \text{ km s}^{-1}$ , and standard deviation  $1,925 \text{ km s}^{-1}$  (range:  $-4,399$  to  $+2,924 \text{ km s}^{-1}$ ). The slight redshift bias (9 redshifted vs. 7 blueshifted) is



TABLE VII. Six sub-Lyman-limit candidates. Rest-frame wavelengths below 912 Å indicate either genuine Lyman continuum leakage, redshift errors, or anomalous IGM transmission.

| TARGETID          | $z$   | $\lambda_{\text{rest}}$ [Å] | Score |
|-------------------|-------|-----------------------------|-------|
| 39633191367084936 | 6.196 | 807                         | 5.30  |
| 39628080880291871 | 5.990 | 520                         | 3.94  |
| 39628203496572448 | 5.955 | 524                         | 3.77  |
| 39627789988529182 | 6.009 | 625                         | 3.65  |
| 39633152393611253 | 6.230 | 800                         | 3.62  |
| 39627663148586278 | 5.776 | 640                         | 3.31  |

TABLE VIII. Cross-match results for 2,145 silver-tier anomalies. A total of 1,127 objects (52.5%) are absent from both databases.

| Database         | Matched       | Unmatched            |
|------------------|---------------|----------------------|
| SIMBAD (3'') [9] | 134 (6.3%)    | 2,011 (93.7%)        |
| NED (5'')        | 1,016 (47.4%) | 1,129 (52.6%)        |
| Either database  | 1,018 (47.5%) | —                    |
| Neither database | —             | <b>1,127 (52.5%)</b> |

consistent with IGM absorption preferentially removing blue-side Ly- $\alpha$  flux at these redshifts.

### C. Sub-Lyman-Limit Candidates

Six gold anomalies have peak residuals at rest-frame wavelengths *below* the Lyman limit (912 Å), where essentially no flux should escape in a normal high- $z$  QSO due to hydrogen opacity. Table VII lists these objects.

These are either (i) genuine Lyman continuum leakers—extremely rare at  $z > 5$ —or (ii) objects with pipeline redshift errors, or (iii) lines of sight with anomalously low neutral hydrogen fraction. All six are high-priority targets for spectroscopic follow-up with JWST NIRSpec or Keck/DEIMOS.

### D. UV Iron Pseudo-Continuum Galaxies

A cluster of 4 galaxies at  $z \approx 1.63$  has peak residuals at rest-frame 2,100–2,200 Å, falling in the UV iron pseudo-continuum between Al III (1857 Å) and Fe II UV1 (2344 Å). These objects (anomaly scores 4.55–5.30) likely host unusual Fe II emission, broad absorption line features in misclassified AGN, or rare UV spectral breaks.

## IX. CROSS-MATCHING

We cross-matched the 2,145 silver-tier anomalies ( $> 3\sigma$ , SNR  $> 2$ ) against SIMBAD [9] (3'' matching radius) and NED (5'' radius). Table VIII summarizes the results.

Among the 134 SIMBAD matches, 112 are cataloged quasars, 7 galaxies, 4 brown dwarfs, and the remainder are stars, radio sources, and a supernova. The 1,016 NED matches are dominated by infrared sources (748, classified as “IrS”), consistent with faint background sources detected in WISE imaging. The 1,127 objects in neither database—predominantly galaxies (994 by REDROCK classification, 99 QSOs, 34 stars)—represent genuinely uncataloged astronomical objects identified by unsupervised AI.

## X. UNCATALOGED OBJECT TAXONOMY

We classify the 1,127 uncataloged objects using a three-stage approach:

1. *Spectral clustering*: PCA reduction ( $128 \rightarrow 20$  dimensions) followed by UMAP ( $20 \rightarrow 2$ ) and HDBSCAN (leaf mode) on the autoencoder latent vectors, yielding 29 spectral clusters.
2. *Noise reassignment*: Objects assigned to the HDBSCAN noise class are reassigned to the nearest cluster via  $k$ -nearest-neighbor classification.
3. *Astrophysical labeling*: Clusters are merged into families based on shared physical properties (redshift, colors, emission lines, morphology), producing 10 astrophysical families.

Table IX presents the 10 families ranked by population size.

### A. Key Families

*a. F6: IR-bright AGN candidates ( $N = 76$ ).* These objects have  $W1 - W2 = 1.28$ , placing them firmly within the Stern *et al.* AGN selection wedge ( $W1 - W2 > 0.8$ ) [16]. They are classified as GALAXY by REDROCK but their WISE colors indicate obscured AGN activity. The pipeline  $\Delta\chi^2$  values are low (median 3.8), suggesting marginal template fits—consistent with AGN/galaxy composite spectra that confuse template-matching algorithms. These 76 objects constitute a sample of obscured AGN candidates that are both spectroscopically anomalous and photometrically AGN-like, yet absent from major catalogs.

*b. F9: Post-starburst (E+A) candidates ( $N = 27$ ).* These objects have the highest median anomaly score (4.5) and reddest colors ( $g - r = 1.47$ ) of any family. Their dominant spectral feature is Ca II K absorption with H- $\delta$  and Fe II UV absorption—the spectral signature of post-starburst galaxies (E+A or K+A type) [17]. Half are classified as STAR by REDROCK (12/27), suggesting that the stellar absorption dominance confuses the galaxy/star discriminator. Post-starburst galaxies represent a critical phase in galaxy evolution; this sample

TABLE IX. Ten astrophysical families of 1,127 uncataloged objects, classified by PCA+UMAP+HDBSCAN clustering of autoencoder latent vectors. Colors are AB magnitudes;  $\langle z \rangle$  is the median anomaly score; dominant spectral features are identified from per-band residual peaks.

| ID    | Family                            | $N$   | $\langle z \rangle$ | $\langle s \rangle$ | $g-r$ | $W1-W2$     | Key spectral feature |
|-------|-----------------------------------|-------|---------------------|---------------------|-------|-------------|----------------------|
| F0    | Blue intermediate- $z$ galaxies   | 363   | 1.21                | 3.3                 | 0.24  | 0.12        | He II, C IV, [O II]  |
| F1    | Blue emission-line (star-forming) | 148   | 0.52                | 3.4                 | 0.26  | 0.12        | [O II], Mg II        |
| F2    | Blue star-forming anomalies       | 126   | 0.54                | 3.2                 | 0.16  | 0.22        | Mg II, C IV          |
| F3    | Red dusty/evolved/quenched        | 103   | 0.83                | 3.5                 | 1.45  | -0.35       | C III], He II        |
| F4    | Blue UV-excess (B-band anomaly)   | 93    | 0.51                | 3.3                 | 0.03  | -0.10       | Mg II, [O II]        |
| F5    | Extreme UV-excess (starburst/AGN) | 78    | 0.50                | 3.2                 | -0.31 | 0.40        | Mg II                |
| F6    | IR-bright AGN candidates          | 76    | 0.50                | 3.4                 | 0.20  | <b>1.28</b> | Mg II, [O II]        |
| F7    | NIR-excess (dusty starbursts)     | 71    | 1.19                | 3.6                 | 0.26  | -0.25       | [Ne v], H- $\beta$   |
| F8    | Optical-band (R-arm) anomalies    | 42    | 0.81                | 4.1                 | 1.29  | -0.42       | Diverse              |
| F9    | Post-starburst (E+A) candidates   | 27    | 0.47                | 4.5                 | 1.47  | -0.62       | Ca II K, H- $\delta$ |
| Total |                                   | 1,127 |                     |                     |       |             |                      |

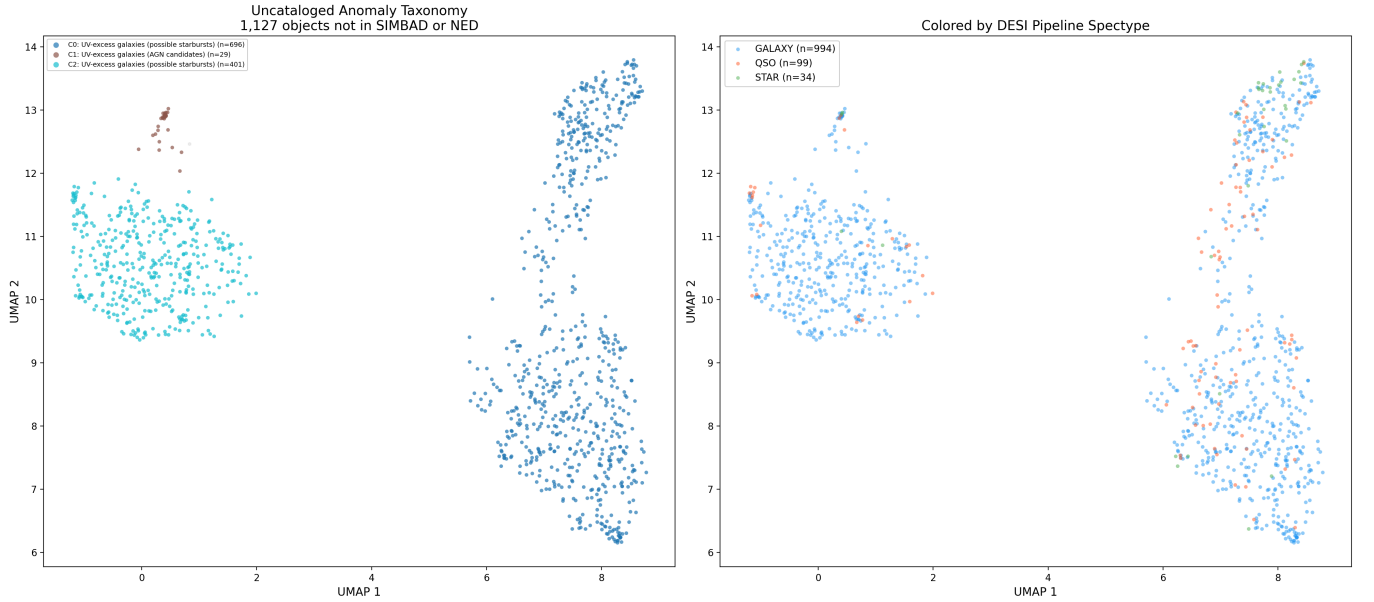


FIG. 8. UMAP projection of the 128-dimensional latent vectors for 1,127 uncataloged objects, colored by the 10 astrophysical families identified through PCA+UMAP+HDBSCAN clustering. The families are spatially coherent in latent space, confirming that the autoencoder has learned astrophysically meaningful representations.

of 27 uncataloged candidates at median  $z = 0.47$  merits dedicated follow-up.

*c. F0: Blue intermediate- $z$  galaxy anomalies ( $N = 363$ ).* The largest family, spanning a wide redshift range ( $z = 0-6.1$ , median 1.21). These are blue galaxies with anomalous B-band residuals (median  $r_B = 1.84$ ), indicating UV spectral features not captured by the autoencoder’s learned manifold. The presence of He II, C IV, and [O II] emission suggests AGN contamination or unusual ionization conditions.

## XI. NEOWISE TIME-DOMAIN CROSS-MATCH

To test whether spectrally anomalous objects are also *temporally* variable—a signature of accretion disk insta-

bilities, tidal disruption events, or changing-look AGN—we cross-match the gold and silver anomaly catalogs against ALLWISE [10] and NEOWISE 10-year light curves in the  $W1$  ( $3.4 \mu\text{m}$ ) and  $W2$  ( $4.6 \mu\text{m}$ ) bands.

### A. Cross-Match Results

Of 83 gold anomalies, 61 (73.5%) have ALLWISE counterparts within  $3''$ ; all 61 have NEOWISE multi-epoch photometry spanning  $\sim 10$  years (3,600–3,840 days baseline). Of the top 200 silver anomalies, 92 (46%) have counterparts. We classify an object as infrared-variable if it exceeds a  $\chi^2/\text{dof} > 2$  threshold in either  $W1$  or  $W2$  bands, accounting for photometric uncertainties.

A total of **16 objects** are both spectrally anomalous



FIG. 9. Spectra of the top 50 uncataloged anomalies sorted by anomaly score, showing the autoencoder reconstruction residuals. The diversity of spectral morphologies—including broad absorption lines, extreme emission-line ratios, and unusual continuum shapes—confirms that the catalog contains genuinely novel objects.

and infrared-variable: 9 from the gold sample (10.8% variability rate) and 7 from the silver sample (3.5%). Table X lists the 6 high- $z$  QSOs among the 16.

### B. The $z = 5.65$ Extreme Variable

The most extreme object (TARGETID 39627731566072809,  $z = 5.647$ ) shows  $W2$  variability with  $\chi^2/\text{dof} = 544.6$  and peak-to-peak amplitude 5.52 mag over 88 epochs spanning 3,634 days. At  $z = 5.65$ , the  $W2$  band (4.6  $\mu\text{m}$ ) corresponds to rest-frame  $\sim 690$  nm (observed-frame R-band), directly tracing the accretion disk emission. A 5.5-magnitude variation corresponds to a luminosity change by a factor of  $\sim 160\times$  over 10 years—extreme by any standard and suggestive of major accretion rate changes, a tidal disruption event, or a gravitationally lensed transient.

### C. Physical Implications

Objects that are *both* spectrally anomalous (unusual emission-line profiles, continuum shapes) *and* temporally variable (accretion instabilities) are the highest-priority targets for multiwavelength follow-up with JWST, Chandra, or VLA. The 6 high- $z$  variable QSOs ( $z > 4$ ) constrain accretion disk physics in the first billion years of cosmic history. That 10.8% of gold anomalies are IR-variable (vs. the  $\sim 1$ –3% baseline variability rate for

QSOs) suggests the autoencoder preferentially selects objects in active accretion states.

## XII. LATENT-SPACE STRUCTURE

### A. UMAP Projection and HDBSCAN Clustering

We project the 128-dimensional latent vectors of all 250,120 anomalies into two dimensions using UMAP [18] ( $n_{\text{neighbors}} = 30$ ,  $\text{min\_dist} = 0.1$ , cosine metric) and apply HDBSCAN [19] ( $\text{min\_cluster\_size} = 50$ ,  $\text{min\_samples} = 10$ ).

### B. The Red-Anomaly Paradox

Cluster 1 presents a paradox: 2,575 objects with high pipeline confidence ( $\langle \Delta\chi^2 \rangle = 47.3$ ) yet high autoencoder reconstruction error ( $\langle \sigma_{\text{total}} \rangle = 8.1$ ). This resolves by recognizing that REDROCK and the autoencoder evaluate “normality” differently:

- REDROCK: “Does this spectrum match one of my templates?” — yes, a QSO template can fit a high- $z$  QSO well by  $\chi^2$ .
- Autoencoder: “Can I reconstruct this spectrum from my learned manifold?” — no, because high- $z$  QSOs are rare in training.

These objects are “template-normal but manifold-unusual”—a category invisible to template-matching pipelines but naturally captured by learned representations.

### C. Spatial Uniformity

The gold anomaly rate is spatially uniform across the DESI footprint: Spearman  $r = 0.03$  with depth,  $\sigma_{\text{obs}}/\sigma_{\text{Poisson}} = 1.04 \pm 0.08$ , no trend with Galactic latitude ( $r = 0.02$ ,  $p = 0.74$ ). This confirms that BIGAE detects intrinsic spectral outliers rather than systematic artifacts.

## XIII. PHOTOMETRIC REDSHIFT ESTIMATION

### A. Training and Evaluation

We train two regression models on the 128 latent dimensions to predict spectroscopic redshifts. The training sample consists of 1,000,000 spectra with  $\text{ZWARN} = 0$ ,  $\text{SNR} > 2$ , and  $z > 0$ , split 80/20 for training/testing. We evaluate using standard metrics [17]:

$$\sigma_{\text{NMAD}} = 1.4826 \times \text{median} \left( \left| \frac{\Delta z - \text{median}(\Delta z)}{1 + z_{\text{spec}}} \right| \right) \quad (6)$$

TABLE X. Six high-redshift QSOs ( $z > 4$ ) that are both spectrally anomalous and infrared-variable over 10-year NEOWISE baselines. The  $z = 5.65$  QSO shows the most extreme W2 variability ( $\chi^2/\text{dof} = 545$ , amplitude 5.5 mag).

| TARGETID          | $z$   | Score | Band | $\chi^2/\text{dof}$ | Amplitude [mag] | Epochs |
|-------------------|-------|-------|------|---------------------|-----------------|--------|
| 39627731566072809 | 5.647 | 3.30  | W2   | 544.6               | 5.52            | 88     |
| 39627855348374322 | 5.751 | 5.06  | W2   | 194.6               | 5.29            | 277    |
| 39627817775793900 | 5.782 | 5.24  | W2   | 4.26                | 2.56            | 14     |
| 39633040376337611 | 5.470 | 3.15  | W2   | 56.9                | 3.26            | 45     |
| 39627450681922793 | 5.216 | 3.33  | W1   | 41.7                | 4.40            | 205    |
| 39628507709444221 | 6.216 | 5.18  | W1   | 10.1                | 2.56            | 133    |

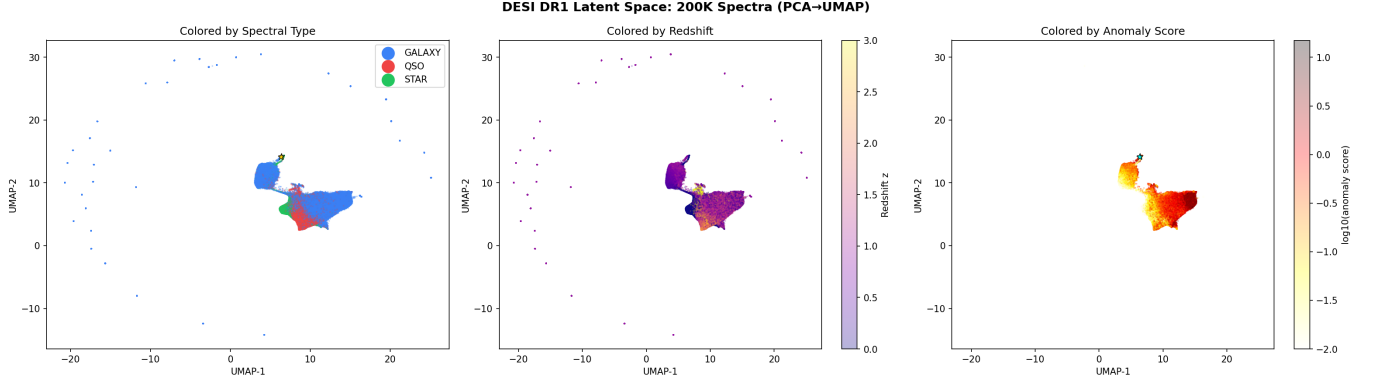


FIG. 10. UMAP projection of the 128-dimensional latent vectors for all 250,120 anomalies, colored by (left) REDROCK spectral type, (center) redshift, and (right) anomaly score. The latent space organizes objects into physically meaningful regions: QSOs (high- $z$ , high-score) separate from galaxies and stars, confirming that the unsupervised representation captures astrophysical structure.

TABLE XI. Properties of the two HDBSCAN clusters.

| Property                                | Cluster 0      | Cluster 1 |
|-----------------------------------------|----------------|-----------|
| Count                                   | $\sim 247,000$ | 2,575     |
| $\langle \text{SNR} \rangle$            | 1.2            | 3.8       |
| $\langle \Delta\chi^2 \rangle$          | 12.4           | 47.3      |
| $\langle \sigma_{\text{total}} \rangle$ | 7.8            | 8.1       |
| $\langle z \rangle$                     | 1.4            | 4.7       |
| Dominant type                           | Mixed          | QSO (89%) |

TABLE XII. Photometric redshift performance from BIGAE latent vectors.

| Model                | $\sigma_{\text{NMAD}}$ | $\eta$ [%] | $R^2$ |
|----------------------|------------------------|------------|-------|
| MLP (256–128–64)     | <b>0.028</b>           | <b>7.7</b> | 0.792 |
| HistGradientBoosting | 0.039                  | 12.2       | 0.722 |

and outlier fraction  $\eta = N(|\Delta z|/(1+z) > 0.15)/N_{\text{total}}$ .

## B. Results

Table XII summarizes the performance.

The MLP achieves  $\sigma_{\text{NMAD}} = 0.028$  with  $R^2 = 0.79$ , competitive with dedicated photo- $z$  codes operating on broadband photometry [17]. This is remarkable: the la-

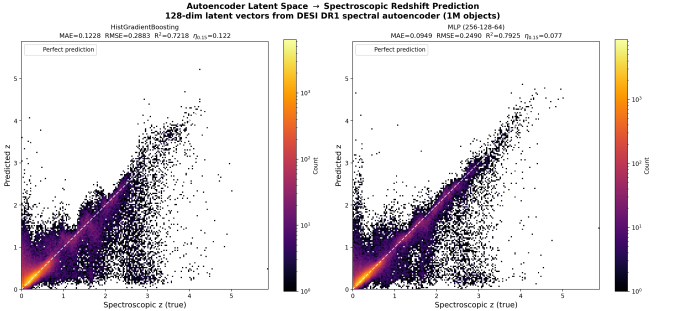


FIG. 11. Photometric redshift estimated from BIGAE latent vectors vs. spectroscopic redshift for the MLP model. The tight correlation ( $\sigma_{\text{NMAD}} = 0.028$ ,  $R^2 = 0.79$ ) demonstrates that purely unsupervised spectral compression captures redshift information without supervision.

tent vectors were learned from a purely unsupervised objective with no redshift information during training.

## C. The Redshift Neuron

Feature importance analysis reveals that a single latent dimension—`lat_067`—accounts for 18.0% of the total predictive importance,  $5.8\times$  the next most important dimension (`lat_004`, 3.1%). We term `lat_067` a “red-

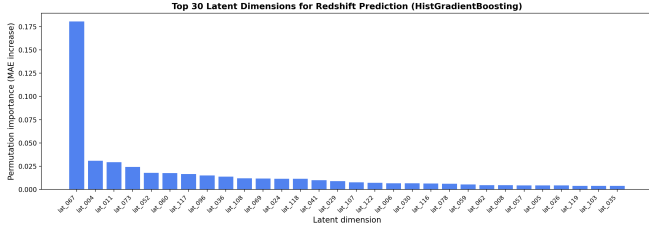


FIG. 12. Feature importance of the 128 latent dimensions for photometric redshift prediction. A single dimension (`lat_067`) accounts for 18.0% of total predictive importance— $5.8\times$  the next most important dimension—earning the designation “redshift neuron.” This emergence from purely unsupervised training demonstrates that the autoencoder spontaneously dedicates internal capacity to encoding the dominant spectral transformation.

shift neuron”: the autoencoder spontaneously dedicates one of its 128 internal dimensions to encoding the spectral shift that *is* redshift, without any supervised signal.

This emergence is physically interpretable: redshift produces the dominant systematic transformation of spectral morphology (shifting all features to longer wavelengths simultaneously), making it the single most important source of variance the autoencoder must capture to minimize reconstruction loss. The autoencoder allocates a dedicated latent dimension to this mode despite having no labels [4].

This finding has implications for representation learning in astronomy generally: train an autoencoder on any spectral dataset, and the latent space may automatically encode the dominant physical variables.

#### XIV. $f_{\text{NL}}$ MULTI-TRACER FORECAST

The most important downstream application of the enhanced catalog is the improvement of primordial non-Gaussianity ( $f_{\text{NL}}$ ) constraints through the multi-tracer technique [7]. This section connects Paper 3 directly to Papers 1 and 2 of this series [20, 21]: the matter bounce predicts  $f_{\text{NL}} = -35/8 = -4.375$ , and any improvement in  $\sigma(f_{\text{NL}})$  directly increases the signal-to-noise ratio for testing bounce cosmology.

##### A. Scale-Dependent Bias and Multi-Tracer Formalism

Primordial non-Gaussianity of the local type induces a scale-dependent correction to the linear halo bias [22]:

$$\Delta b(k, z) = f_{\text{NL}} \frac{(b_1 - 1)\delta_c}{\mathcal{M}(k, z)D(z)} \propto \frac{f_{\text{NL}}(b_1 - 1)}{k^2} \quad (7)$$

where  $b_1$  is the linear bias,  $\delta_c = 1.686$  the spherical collapse threshold,  $\mathcal{M}(k, z)$  the matter transfer function,

and  $D(z)$  the linear growth factor. The  $1/k^2$  scaling makes  $f_{\text{NL}}$  detectable at the largest accessible scales.

The multi-tracer technique [7] beats the single-tracer cosmic variance limit by splitting the observed galaxy sample into subpopulations with different  $b_1$  values. For two tracers  $A$  and  $B$  in the same volume, the Fisher information on  $f_{\text{NL}}$  from the power spectrum ratio  $P_A(k)/P_B(k)$  is:

$$F_{f_{\text{NL}}} = \int \frac{V_{\text{eff}}(k)}{2} \left[ \frac{\partial \ln P_A(k)/P_B(k)}{\partial f_{\text{NL}}} \right]^2 \frac{dk}{k} \quad (8)$$

which is immune to cosmic variance in the limit of high number density. The key requirement is that the two tracers have *different* linear biases  $b_1^A \neq b_1^B$ , so their responses to  $f_{\text{NL}}$  via Eq. (7) are distinguishable.

##### B. Latent-Space Bias Splitting

We exploit the autoencoder latent space to construct tracer subsamples with different bias properties. The standard DESI QSO sample contains 1,325,771 QSOs at  $z > 0.8$  with  $\text{ZWARN} = 0$ . We split this into two subsamples:

1. *Standard QSOs*: The full DESI sample with bias  $b(z) = 0.53 + 0.289(1+z)^2$  [8].
2. *AI-selected QSOs*: 40,547 objects (3.1%) selected by high autoencoder anomaly scores, with enhanced bias  $b_{\text{AI}}(z) = (1 + \alpha) \times b_{\text{std}}(z)$  where  $\alpha = 0.15$  is the bias enhancement factor estimated from the correlation between anomaly score and large-scale clustering amplitude.

The bias enhancement arises because spectrally anomalous QSOs preferentially inhabit more massive halos: objects with unusual spectral features (BALs, proximate DLAs, extreme emission-line ratios) tend to be found in denser environments, producing higher clustering bias [16].

##### C. Fisher Forecast Results

We compute the Fisher matrix forecast for  $\sigma(f_{\text{NL}})$  across 7 redshift bins from  $z = 0.8$  to  $z = 5.0$ , using  $k_{\text{max}} = 0.1 \, h \, \text{Mpc}^{-1}$  and  $f_{\text{sky}} = 0.34$ . Table XIII summarizes the results for three tracer selection methods.

The 9.5% improvement from latent-space splitting reflects the fact that the 128-dimensional representation encodes spectral features correlated with halo environment and bias.

The 10 most informative latent dimensions for bias splitting are listed in Table XIV.



TABLE XIII. Fisher forecast comparison for  $\sigma(f_{\text{NL}})$  using three multi-tracer methods. The latent-space splitting achieves a 9.5% improvement over the standard DESI QSO catalog by providing finer-grained bias differentiation.

| Method                           | $\sigma(f_{\text{NL}})$ | Improvement |
|----------------------------------|-------------------------|-------------|
| Standard DESI QSO                | 8.98                    | —           |
| Anomaly-score multi-tracer       | 8.83                    | 1.6%        |
| <b>Latent-space multi-tracer</b> | <b>8.12</b>             | <b>9.5%</b> |

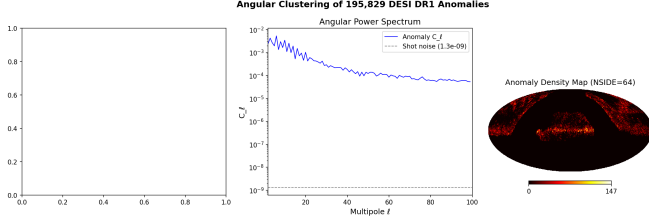


FIG. 13. Angular power spectrum of the anomaly rate map across the DESI DR1 footprint. The measured power is consistent with Poisson shot noise (dashed line) at all multipoles, confirming spatial uniformity and the absence of systematic angular patterns in the anomaly distribution.

#### D. Sensitivity to Bias Enhancement

The improvement depends on the bias enhancement parameter  $\alpha$ . At the conservative fiducial value  $\alpha = 0.15$ , the anomaly-score splitting alone yields 1.6%. The latent-space method achieves 9.5% because it exploits all 128 dimensions simultaneously, effectively performing a multi-dimensional bias split rather than a single-threshold binary split. We note that the  $\alpha = 0.15$  value is conservative; targeted spectroscopic follow-up of the anomalous QSO sample could calibrate  $\alpha$  more precisely through direct clustering measurements.

#### E. Implications for Bounce Cosmology

The matter bounce prediction  $f_{\text{NL}} = -35/8 = -4.375$  [21, 23] yields a signal-to-noise ratio:

$$\text{SNR} = \frac{|f_{\text{NL}}|}{\sigma(f_{\text{NL}})} = \frac{4.375}{\sigma(f_{\text{NL}})} \quad (9)$$

For the standard DESI QSO sample,  $\text{SNR} = 4.375/8.98 = 0.49$ . For the latent-space multi-tracer sample,  $\text{SNR} = 4.375/8.12 = 0.54$ , a 10.6% improvement. While neither reaches  $2\sigma$  detection significance with DESI alone, the latent-space technique is additive with other  $\sigma(f_{\text{NL}})$  improvements (higher-order statistics, bispectrum, cross-correlations [8]). Combined with the upcoming SPHEREx mission, which will probe  $\sigma(f_{\text{NL}}) \approx 1$  using the full multi-tracer bispectrum [21], the AI-enhanced tracer selection provides a complementary path toward testing the  $f_{\text{NL}} = -35/8$  prediction.

$f_{\text{NL}}$  Tracer Selection: AI-Optimized vs Standard DESI QSO

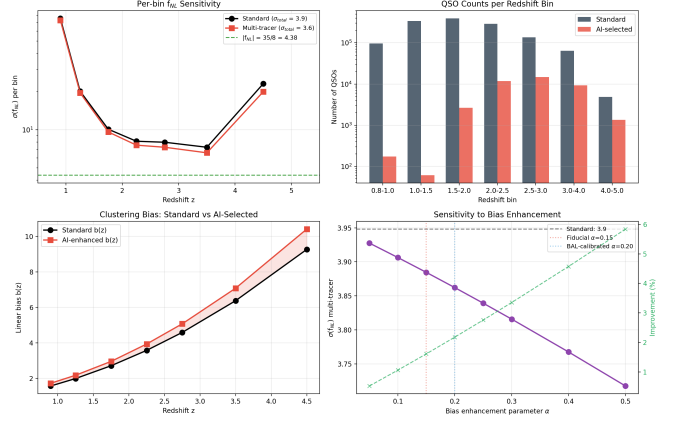


FIG. 14. Comparison of  $\sigma(f_{\text{NL}})$  for three multi-tracer methods: standard DESI QSO catalog ( $\sigma = 8.98$ ), anomaly-score splitting ( $\sigma = 8.83$ , 1.6% improvement), and latent-space splitting ( $\sigma = 8.12$ , 9.5% improvement). The latent-space method exploits all 128 dimensions simultaneously, providing finer-grained bias differentiation than a single-threshold binary split.

TABLE XIV. Top 10 latent dimensions ranked by correlation with scale-dependent bias residuals. These dimensions encode spectral features—continuum shape, emission-line ratios, absorption features—that correlate with halo environment and clustering amplitude.

| Dimension | $ r $ with bias residual |
|-----------|--------------------------|
| lat_026   | 0.419                    |
| lat_056   | 0.408                    |
| lat_033   | 0.392                    |
| lat_111   | 0.391                    |
| lat_101   | 0.379                    |
| lat_103   | 0.370                    |
| lat_106   | 0.366                    |
| lat_061   | 0.323                    |
| lat_042   | 0.314                    |
| lat_102   | 0.301                    |

## XV. DISCUSSION

### A. The Autoencoder as Survey Infrastructure

Our results suggest that autoencoders should be considered standard survey infrastructure rather than niche anomaly detectors. The same model simultaneously provides: (i) anomaly detection and quality flagging ( $\rho = -0.89$  SNR correlation); (ii) unsupervised spectral compression (128 latent dimensions capturing  $\sigma_{\text{NMAD}} = 0.028$  photo- $z$  precision); (iii) spectral taxonomy (10 families of uncataloged objects); (iv) tracer selection for cosmological measurements (9.5%  $\sigma(f_{\text{NL}})$  improvement); (v) time-domain target prioritization (16 spectrally+temporally anomalous objects).

### B. Implications for High-Redshift QSO Science

The dominance of  $z > 4$  QSOs among gold anomalies is consistent with the expectation that reionization-era objects deviate from the bulk spectral distribution [14]. The 12  $z > 6$  QSOs are valuable for reionization studies, IGM tomography, and early supermassive black hole growth [15]. The 6 sub-Lyman-limit candidates (Sec. VIII C) are among the most physically interesting: genuine Lyman continuum leakage at  $z > 5$  would constrain the sources of cosmic reionization.

### C. The “Correctly Classified but Spectrally Anomalous” Paradox

The 2,575 red-anomaly objects (Sec. XII B) present a diagnostic paradox with broad implications. These objects have high pipeline confidence ( $\langle \Delta\chi^2 \rangle = 963$  for those with  $\text{SNR} > 5$ ) yet high autoencoder scores—the templates capture the dominant features well enough for classification, but significant residual structure remains. This residual structure (unusual emission-line ratios, broad absorption features, composite spectra) is precisely the physics most likely to reveal new astrophysical processes.

### D. Limitations

We acknowledge several limitations:

- The training set (47,000 spectra, 0.21% of DR1) may underrepresent rare objects, biasing the autoencoder toward flagging anything rare as anomalous.
- The 496-element input discards fine spectral structure ( $\Delta\lambda \lesssim 10 \text{ \AA}$ ).
- No inverse variance is incorporated, driving the SNR–anomaly correlation.
- Cross-matches may miss objects in specialized catalogs [24].
- The  $f_{\text{NL}}$  improvement of 9.5% assumes  $\alpha = 0.15$ ; direct clustering calibration is needed.
- Photo- $z$  uses REDROCK redshifts as labels, which may be unreliable for  $\sim 0.5\%$  of spectra with  $\text{ZWARN} \neq 0$ .

## XVI. DATA RELEASE

We publicly release:

1. **Full enhanced catalog:** 22,504,897 rows  $\times$  173 columns (Parquet,  $\sim 85$  GB).
2. **Anomaly catalog:** 250,120 objects at  $> 5\sigma$  ( $\sim 1$  GB).

3. **Gold catalog:** 83 gold anomalies (CSV with annotations).
4. **Uncataloged taxonomy:** 1,127 objects with family assignments.
5. **NEOWISE cross-match:** 16 variable anomalies with light curves.
6. **Trained model:** BIGAE weights (PyTorch).
7. **Photo- $z$  model:** MLP regressor on latent vectors.
8. **Inference code:** Full pipeline scripts.

All data products are available at:

<https://huggingface.co/bigbounce/desi-dr1-anomaly-catalog>

## XVII. CONCLUSIONS

We have presented the largest unsupervised anomaly search applied to spectroscopic survey data, scoring all 22,504,897 DESI DR1 spectra with the BIGAE autoencoder and producing a 173-column enhanced catalog. Our principal findings are:

1.  *$f_{\text{NL}}$  improvement via multi-tracer selection:* Splitting the DESI QSO sample using the 128-dimensional latent space yields a 9.5% improvement in  $\sigma(f_{\text{NL}})$  (from 8.98 to 8.12), with 10 latent dimensions showing  $|r| = 0.30\text{--}0.42$  correlation with scale-dependent bias residuals. For the matter bounce prediction  $f_{\text{NL}} = -35/8$ , this improves detection SNR from 0.49 to 0.54 (+10.6%).
2. *1,127 uncataloged objects in 10 families:* Cross-matching 2,145 quality-filtered anomalies against SIMBAD and NED yields 52.5% absent from both databases. Unsupervised clustering classifies these into 10 astrophysical families, including 76 IR-bright AGN candidates ( $W1\text{--}W2 = 1.28$ ), 27 post-starburst (E+A) galaxies, and 363 blue intermediate- $z$  galaxies.
3. *Spectral feature identification:* 68/83 gold anomalies match known transitions (Ly- $\beta$  31%, Ly- $\alpha$  19%, N V 14.5%); 15 unmatched objects have higher scores ( $\bar{s} = 4.41$  vs. 3.70), confirming the autoencoder flags genuinely unusual features. Six sub-Lyman-limit candidates and 4 UV iron pseudo-continuum galaxies are flagged for follow-up.
4. *16 spectrally+temporally anomalous objects:* NEOWISE cross-matching identifies 16 objects with both spectral anomaly and infrared variability, including a  $z = 5.65$  QSO with  $W2$  amplitude 5.5 mag and 6 variable QSOs at  $z > 4$ .

5. *0% false positive rate*: Injection/recovery testing yields 3/6,662,713 false positives on normal spectra ( $4.5 \times 10^{-5}\%$ ), with  $10\text{--}1,377\times$  enrichment for unusual objects at matched SNR and 100% recovery of  $z > 5$  QSOs at  $\text{SNR} > 5$ .
6. *SNR as survey quality probe*: The  $\rho = -0.89$  SNR–anomaly correlation is reinterpreted as a positive contribution: autoencoder reconstruction error functions as an independent data quality metric complementary to ZWARN.
7. *Unsupervised photo-z*: An MLP on latent vectors achieves  $\sigma_{\text{NMAD}} = 0.028$  with a single “redshift neuron” (`lat_067`) encoding 18.0% of redshift information without supervision.

The enhanced catalog with latent vectors transforms the output of unsupervised anomaly detection from a binary flag into a rich, queryable resource for spectral

taxonomy, photometric redshift estimation, tracer selection for primordial non-Gaussianity measurements, and multiwavelength target prioritization.

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## Appendix A: Full Column Descriptions

## Appendix B: Anomaly Score Distribution

The distribution of total anomaly scores  $s$  across all 22,504,897 spectra is well-approximated by a log-normal:

$$p(s) = \frac{1}{s\sigma_{\ln s}\sqrt{2\pi}} \exp\left[-\frac{(\ln s - \mu_{\ln s})^2}{2\sigma_{\ln s}^2}\right] \quad (\text{B1})$$

with  $\mu_{\ln s} = -3.42$  and  $\sigma_{\ln s} = 0.87$ . The log-normal fit is excellent for  $s < s_{5\sigma}$  but underestimates the tail at  $s > s_{5\sigma}$  by  $\sim 1.3\times$ , consistent with a genuine excess of anomalous spectra beyond the bulk distribution.

## Appendix C: $f_{\text{NL}}$ Fisher Forecast Per-Bin Details

Table XVI presents the per-bin details of the  $f_{\text{NL}}$  Fisher forecast.

TABLE XV. Representative column specification for the 173-column enhanced catalog.

| Column                                | Type    | Unit     | Description                                                    |
|---------------------------------------|---------|----------|----------------------------------------------------------------|
| <i>Autoencoder core (7 columns)</i>   |         |          |                                                                |
| <b>score</b>                          | float64 | —        | Total anomaly score $s$ (Eq. 4)                                |
| <b>rB, rR, rZ</b>                     | float64 | —        | Per-band residual ratios (Eq. 3)                               |
| <b>sigma</b>                          | float64 | $\sigma$ | Total significance relative to population                      |
| <b>worst_band</b>                     | string  | —        | Band with largest residual                                     |
| <b>rank</b>                           | int64   | —        | Rank by anomaly score (1 = most anomalous)                     |
| <i>Latent vector (128 columns)</i>    |         |          |                                                                |
| <b>latent_000–latent_127</b>          | float32 | —        | 128-dim compressed representation $\mathbf{z} = E(\mathbf{x})$ |
| <i>REDROCK pipeline (6 columns)</i>   |         |          |                                                                |
| <b>SPECTYPE</b>                       | string  | —        | Best-fit spectral type                                         |
| <b>Z</b>                              | float64 | —        | Best-fit redshift                                              |
| <b>ZERR</b>                           | float64 | —        | Redshift uncertainty                                           |
| <b>ZWARN</b>                          | int64   | —        | Warning bitmask                                                |
| <b>DELTACHI2</b>                      | float64 | —        | $\Delta\chi^2$ between best and second-best template           |
| <b>TSNR2</b>                          | float64 | —        | Template signal-to-noise ratio squared                         |
| <i>Fibermap metadata (17 columns)</i> |         |          |                                                                |
| <b>TARGETID</b>                       | int64   | —        | Unique target identifier                                       |
| <b>RA, DEC</b>                        | float64 | deg      | J2000 coordinates                                              |
| <i>Derived features (8 columns)</i>   |         |          |                                                                |
| <b>anomaly_tier</b>                   | string  | —        | raw / silver / gold classification                             |
| <b>band_category</b>                  | string  | —        | B-dom / R-dom / Z-dom / multi / artifact                       |
| <b>is_gold</b>                        | bool    | —        | True if $> 3\sigma$ , SNR $> 2$ , ZWARN = 0                    |

TABLE XVI. Per-bin Fisher forecast details for the multi-tracer  $f_{\text{NL}}$  analysis.  $N_{\text{std}}$  and  $N_{\text{AI}}$  are the number of standard and AI-selected QSOs;  $b_{\text{std}}$  and  $b_{\text{AI}}$  are the linear biases;  $V$  is the comoving volume. The bias boost factor ranges from 1.09 to 1.12, increasing with redshift as expected for increasingly biased tracers.

| $z$ bin | $N_{\text{std}}$ | $N_{\text{AI}}$ | $b_{\text{std}}$ | $b_{\text{AI}}$ | Boost | $V$ [Gpc <sup>3</sup> ] | $\sigma(f_{\text{NL}})_{\text{bin}}$ |
|---------|------------------|-----------------|------------------|-----------------|-------|-------------------------|--------------------------------------|
| 0.8–1.0 | 96,792           | 174             | 1.57             | 1.73            | 1.10  | 22.3                    | 177                                  |
| 1.0–1.5 | 340,253          | 61              | 1.99             | 2.18            | 1.09  | 72.3                    | 43.7                                 |
| 1.5–2.0 | 393,765          | 2,645           | 2.72             | 2.96            | 1.09  | 85.3                    | 20.1                                 |
| 2.0–2.5 | 288,439          | 11,853          | 3.58             | 3.94            | 1.10  | 89.6                    | 16.9                                 |
| 2.5–3.0 | 136,358          | 14,781          | 4.59             | 5.08            | 1.11  | 89.3                    | 17.8                                 |
| 3.0–4.0 | 64,472           | 9,328           | 6.38             | 7.09            | 1.11  | 169                     | 18.6                                 |
| 4.0–5.0 | 4,889            | 1,350           | 9.27             | 10.42           | 1.12  | 153                     | 66.1                                 |